

AM/FM/PM MODULATION ANALYSIS

Specifications

R&S®FSW3-KM107

R&S®FSW-K7

R&S®ESW-K7

R&S®FSWP-K7

R&S®FSMR3-K7

R&S®FSV3-K7

R&S®FPS-K7

R&S®FPL1-K7

R&S®VSE-K7



Specifications
Version 11.00

ROHDE & SCHWARZ

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Definitions

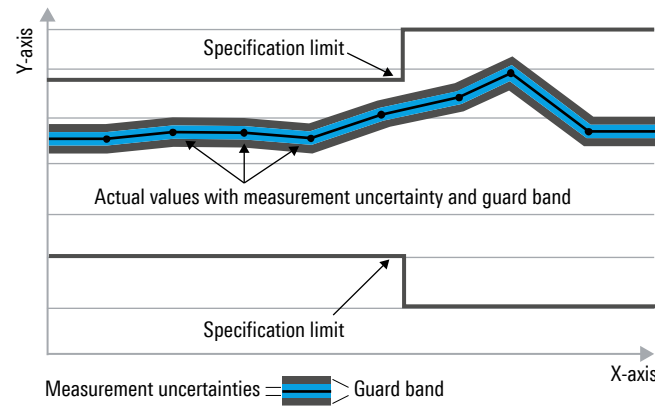
General

Product data applies under the following conditions:

- Three hours of storage at ambient temperature followed by 30 minutes of warm-up operation
- Specified environmental conditions met
- Recommended calibration interval adhered to
- All internal automatic adjustments performed, if applicable

Specifications with limits

Represent warranted product performance by means of a range of values for the specified parameter. These specifications are marked with limiting symbols such as $<$, $\leq$, $>$, $\geq$, $\pm$, or descriptions such as maximum, limit of, minimum. Compliance is ensured by testing or is derived from the design. Test limits are narrowed by guard bands to take into account measurement uncertainties, drift and aging, if applicable.



Non-traceable specifications with limits (n. trc.)

Represent product performance that is specified and tested as described under "Specifications with limits" above. However, product performance in this case cannot be warranted due to the lack of measuring equipment traceable to national metrology standards. In this case, measurements are referenced to standards used in the Rohde & Schwarz laboratories.

Specifications without limits

Represent warranted product performance for the specified parameter. These specifications are not specially marked and represent values with no or negligible deviations from the given value (e.g. dimensions or resolution of a setting parameter). Compliance is ensured by design.

Typical data (typ.)

Characterizes product performance by means of representative information for the given parameter. When marked with $<$, $>$ or as a range, it represents the performance met by approximately 80 % of the instruments at production time. Otherwise, it represents the mean value.

Nominal values (nom.)

Characterize product performance by means of a representative value for the given parameter (e.g. nominal impedance). In contrast to typical data, a statistical evaluation does not take place and the parameter is not tested during production.

Measured values (meas.)

Characterize expected product performance by means of measurement results gained from individual samples.

Uncertainties

Represent limits of measurement uncertainty for a given measurand. Uncertainty is defined with a coverage factor of 2 and has been calculated in line with the rules of the Guide to the Expression of Uncertainty in Measurement (GUM), taking into account environmental conditions, aging, wear and tear.

Device settings and GUI parameters are indicated as follows: "parameter: value".

Non-traceable specifications with limits, typical data as well as nominal and measured values are not warranted by Rohde & Schwarz.

In line with the 3GPP standard, chip rates are specified in million chips per second (Mcps), whereas bit rates and symbol rates are specified in billion bit per second (Gbps), million bit per second (Mbps), thousand bit per second (kbps), million symbols per second (Mps) or thousand symbols per second (kps), and sample rates are specified in million samples per second (Msample/s). Gbps, Mcps, Mbps, Mps, kbps, kps and Msample/s are not SI units.

Specifications

The specifications of the R&S®FSW3-KM107, R&S®Fxx-K7 and R&S®VSE-K7 analog demodulation application are based on the specifications of the corresponding instruments and are not verified during instrument calibration. Measurement uncertainties are given as 95 % confidence intervals. The specified level measurement errors do not take into account systematic errors due to reduced signal to noise ratio (S/N).

General remarks

This specifications document covers the R&S®FSW3-KM107, the R&S®FSW-K7, the R&S®ESW-K7, the R&S®FSWP-K7 ¹, the R&S®FSMR3-K7 ², the R&S®FSV3-K7, the R&S®FPS-K7, the R&S®FPL1-K7 options and the R&S®VSE-K7 software.

The R&S®FSW3-KM107, R&S®FSW-K7, R&S®ESW-K7, R&S®FSWP-K7, R&S®FSMR3-K7, R&S®FSV3-K7, R&S®FPS-K7 and the R&S®FPL1-K7 options are summarized with the term R&S®Fxx-K7. The R&S®Fxx-K7 runs on the analyzer.

The R&S®VSE-K7 runs on a PC that can be connected to the analyzers and oscilloscopes.

If not stated otherwise, the specified values are device specific, e.g. the same value applies to the R&S®FSW-K7 and the R&S®VSE-K7 with a connected FSW.

For better readability, devices and options listed in the tables are written without the prefix "R&S®".

Overview

	FSWX	FSW	ESW/ FSMR3000	FSWP	FSVA3000/ FSV3000	FPS	FPL ³	EPL	RTO
R&S®FSW3-KM107, R&S®Fxx-K7 software that runs on device	• FSW3-KM107	• FSW-K7	• ESW-K7/ FSMR3-K7 ²	• FSWP-K7 ¹	• FSV3-K7	• FPS-K7	• FPL1-K7	• FPL1-K7	–
R&S®VSE-K7 PC software that can be connected to device	•	•	–	•	•	•	•	–	•

¹ Requires R&S®FSWP-B1 option.

² Requires R&S®FSMR3-B1 option.

³ For the FPL, all limits are only valid for RF frequencies of at least 12 MHz.

AM/FM/PM modulation analysis ⁴

		FSWX	FSW	ESW/ FSMR3000	FSWP	FSVA3000/ FSV3000	FPS	FPL ⁵	EPL	RTO
Demodulation bandwidth	100 Hz to 6.4 kHz	•	•	•	•	•	•	•	•	•
	12.5 kHz to 1.6 MHz (binary steps)	•	•	•	•	•	•	•	•	•
	3 MHz	•	•	•	•	•	•	•	•	•
	5 MHz	•	•	•	•	•	•	•	•	•
	8 MHz	•	•	•	•	•	•	•	•	•
	10 MHz	•	•	•	•	•	•	•	•	•
	18 MHz	•	•	•	•	•	•	•	—	•
	28 MHz	•	•	•	•	•	•	•	—	•
	40 MHz	•	•	•	•	•	•	•	—	•
	80 MHz	•	•	•	•	•	•	—	—	•
	160 MHz	•	•	—	•	•	•	—	—	•
	320 MHz	•	•	—	•	—	—	—	—	•
	500 MHz	•	•	—	—	—	—	—	—	•
	1 GHz	•	•	—	—	—	—	—	—	•
	2 GHz	•	•	—	—	—	—	—	—	•
Recording length	maximum	24000001 sample	24000001 sample	24000001 sample	24000001 sample	24000001 sample	24000001 sample	1600001 sample ⁶	1600001 sample	9999900 sample
Recording time	100 Hz	192000 s	196608 s	196608 s	196608 s	196608 s	196608 s	83184 s ⁶	83184 s	9999 s
	6.4 kHz	3000 s	3072 s	3072 s	3072 s	3072 s	3072 s	1299 s ⁶	1299 s	639 s
	12.5 kHz	1536 s	1536 s	1536 s	1536 s	1536 s	1536 s	649 s ⁶	649 s	639 s
	1.6 MHz	12 s	12 s	12 s	12 s	12 s	12 s	5 s ⁶	5 s	4.9 s
	3 MHz	6.4 s	6 s	6 s	6 s	6 s	6 s	2.5 s ⁶	2.5 s	2.6 s
	5 MHz	3.84 s	3 s	3 s	3 s	3 s	3 s	1.2 s ⁶	1.2 s	1.5 s
	8 MHz	2.4 s	1.5 s	1.5 s	1.5 s	1.5 s	1.5 s	634 ms ⁶	634 ms	990 ms
	10 MHz	1.92 s	750 ms	750 ms	750 ms	750 ms	750 ms	317 ms ⁶	317 ms	750 ms
	18 MHz	1.066 s	750 ms	750 ms	750 ms	750 ms	750 ms	317 ms ^{5, 6}	—	440 ms
	28 MHz	685 ms	375 ms	375 ms	375 ms	375 ms	375 ms	158 ms ^{5, 6}	—	280 ms
	40 MHz	480 ms	375 ms	375 ms	375 ms	375 ms	375 ms	158 ms ^{5, 6}	—	190 ms
	80 MHz	240 ms	187.5 ms	187.5 ms	187.5 ms	187.5 ms	187.5 ms	—	—	90 ms
	160 MHz	120 ms	120 ms	—	120 ms	120 ms	120 ms	—	—	45 ms
	320 MHz	60 ms	60 ms	—	60 ms	—	—	—	—	24 ms
	500 MHz	38.4 ms	40 ms	—	—	—	—	—	—	8.3 ms
	1 GHz	19.2 ms	19.2 ms	—	—	—	—	—	—	7.9 ms
	2 GHz	9.6 ms	9.6 ms	—	—	—	—	—	—	3.9 ms

⁴ Depends on the hardware configuration. For details, see FSWX/FSW/R&S®ESW/R&S®FSWP/R&S®FSMR3000/R&S®FSVA3000/R&S®FSV3000/R&S®FPS/FPL/R&S®EPL and R&S®RTO specifications.

⁵ R&S®FPL1-B40 option required for 18/28/40 MHz.

⁶ For FPL with R&S®VSE-K7 option the values as for R&S®FSV3-K7 option apply.

		FSWX	FSW	ESW/ FSMR3000	FSWP	FSVA3000/ FSV3000	FPS	FPL	EPL	RTO
Display	frequency versus time (FM)	•	•	•	•	•	•	•	•	•
	amplitude versus time (AM)	•	•	•	•	•	•	•	•	•
	phase versus time (PM)	•	•	•	•	•	•	•	•	•
	RF power versus time	•	•	•	•	•	•	•	•	•
	RF spectrum (FFT)	•	•	•	•	•	•	•	•	•
	AF spectrum (FFT)	•	•	•	•	•	•	•	•	•
	modulation deviation (peak, RMS)	•	•	•	•	•	•	•	•	•
	modulation frequency	•	•	•	•	•	•	•	•	•
	carrier offset	•	•	•	•	•	•	•	•	•
	carrier power (power of unmodulated carrier)	•	•	•	•	•	•	•	•	•
	THD	•	•	•	•	•	•	•	•	•
	SINAD	•	•	•	•	•	•	•	•	•

AF (modulation frequency)	
Range	max. $0.5 \times$ demodulation bandwidth
Resolution	5 digits
Measurement uncertainty	0.1 %

		FSWX	FSW	ESW/ FSMR3000	FSWP	FSVA3000/ FSV3000	FPS	FPL	EPL	RTO
Lowpass	demodulation bandwidth ≤ 3 MHz									
	3 kHz	•	•	•	•	•	•	•	•	•
	15 kHz	•	•	•	•	•	•	•	•	•
	23 kHz	•	•	•	•	•	•	•	•	•
	150 kHz	•	•	•	•	•	•	•	•	•
	5/10/25 % of demodulation bandwidth	•	•	•	•	•	•	•	•	•
Highpass	demodulation bandwidth ≤ 1.6 MHz									
	20 Hz	•	•	•	•	•	•	•	•	•
	demodulation bandwidth ≤ 3 MHz									
	50 Hz	•	•	•	•	•	•	•	•	•
	demodulation bandwidth ≤ 8 MHz									
	300 Hz	•	•	•	•	•	•	•	•	•
Deemphasis	demodulation bandwidth ≤ 40 MHz									
	25 μ s	•	•	•	•	•	•	•	•	•
	demodulation bandwidth ≤ 18 MHz									
	50/75 μ s	•	•	•	•	•	•	•	•	•
	demodulation bandwidth ≤ 3 MHz									
	750 μ s	•	•	•	•	•	•	•	•	•
Weighting filters	demodulation bandwidth ≤ 3 MHz									
	ITU-T P.53	•	•	•	•	•	•	•	•	•
	demodulation bandwidth ≤ 1.6 MHz									
	ITU-R unweighted	•	•	•	•	•	•	•	•	•
	demodulation bandwidth ≤ 3 MHz									
	ITU-R weighted	•	•	•	•	•	•	•	•	•
	demodulation bandwidth ≤ 800 kHz									
	A weighted	•	•	•	•	•	•	•	•	•

AM demodulation		FSW	ESW/ FSMR3000	FSWP	FSVA3000/ FSV3000	FPS	FPL	EPL	RTO
Measurement range	modulation depth	0 % to 100 %							
Modulation depth uncertainty	AF ≤ 1 MHz	±(0.2 % + 0.001 × measured value) ^{7,8}					±(0.2 % + 1 % of reading)		1 % of reading + residual AM
Residual AM	demodulation bandwidth ≤ 200 kHz, RMS, RF input level ≥ (RF attenuation/dB – 30) dBm R&S®RTO: RF input level ≥ –30 dBm	RF ≤ 8 GHz	RF ≤ 8 GHz	RF ≤ 8 GHz	RF ≤ 4 GHz	RF ≤ 4 GHz	RF ≤ 7.5 GHz	RF ≤ 30 MHz	RF ≤ 4 GHz
		0.03 % ⁷	0.03 % ⁷	0.03 % ⁷	0.1 %	0.1 %	0.1 %	0.1 %	0.2 %
Harmonic distortion		10 Hz ≤ AF ≤ 1 MHz			10 Hz ≤ AF ≤ 100 kHz				10 Hz ≤ AF ≤ 1 MHz
		0.05 % ⁷			0.3 %				0.4 %
FM rejection	AF ≤ 1 MHz, deviation ≤ 1 MHz	AF + deviation ≤ 0.3 × demodulation bandwidth							AF + deviation ≤ 0.3 × demodulation bandwidth
		1 % + residual AM							

⁷ Refer to R&S®RTO column in case the R&S®FSW-B2000 is activated.

⁸ With R&S®VSE and I/Q files the value increases to ±(0.2 % + 0.003 × measured value).

FM demodulation									
		FSW	ESW/ FSMR3000	FSWP	FSVA3000/ FSV3000	FPS	FPL	EPL	RTO
Measurement range	frequency deviation max. $0.5 \times$ demodulation bandwidth								
Deviation uncertainty	AF ≤ 1 MHz	demodulation bandwidth $\geq 3.3 \times (\text{AF} + \text{deviation})$, demodulation bandwidth $\leq 10 \times (\text{AF} + \text{deviation})$					demodulation bandwidth $\geq 3.3 \times (\text{AF} + \text{deviation})$, demodulation bandwidth $\leq 10 \times (\text{AF} + \text{deviation})$		demodulation bandwidth $\geq 3.3 \times (\text{AF} + \text{deviation})$
		$\pm(0.003 \times (\text{AF} + \text{deviation}) + 2 \text{ Hz})^{7,9}$			$\pm(0.01 \times (\text{AF} + \text{deviation}) + 20 \text{ Hz})$		$\pm(0.01 \times (\text{AF} + \text{deviation}) + 20 \text{ Hz})$		0.2 % of reading + residual FM
Residual FM	demodulation bandwidth $\leq 100 \text{ kHz}$, RMS								
	RF input level $\geq (\text{RF attenuation/dB} - 30) \text{ dBm}$								RF input level $\geq -30 \text{ dBm}$
	RF $\leq 30 \text{ MHz}$	–	–	–	–	–	–	130 Hz	–
	RF $\leq 1 \text{ GHz}$	–	–	–	–	–	–	–	30 Hz
	RF $\leq 3 \text{ GHz}$	–	–	–	–	–	–	–	100 Hz
	RF $\leq 7 \text{ GHz}$	–	–	–	65 Hz	130 Hz	130 Hz	–	–
	RF $\leq 8 \text{ GHz}$	10 Hz	10 Hz	10 Hz	–	–	–	–	–
Harmonic distortion	10 Hz $\leq$ AF $\leq 100 \text{ kHz}$, deviation $< 400 \text{ kHz}$	–	–	–	0.3 %	0.3 %	0.3 % ¹⁰	0.3 % ¹⁴	–
	10 Hz $\leq$ AF $\leq 1 \text{ MHz}$, deviation $< 500 \text{ kHz}$	0.1 %	0.1 %	0.1 %	–	–	–	–	0.1 %
AM rejection	100 Hz $\leq$ AF $\leq 1 \text{ kHz}$, modulation depth 50 %	30 Hz + residual FM							

⁹ With R&S®VSE and I/Q files the value increases to $\pm(0.005 \times (AF + \text{deviation}) + 2 \text{ Hz})$.

¹⁰ With demodulation bandwidth > 5 MHz.

PM demodulation		FSW	ESW/ FSMR3000	FSWP	FSVA3000/ FSV3000	FPS	FPL	EPL	RTO
Measurement range	phase deviation	5000 rad, max. $0.5 \times$ demodulation bandwidth/AF					5000 rad, max. $0.5 \times$ demodulation bandwidth/AF		max. $0.5 \times$ demodulation bandwidth/AF
Phase deviation uncertainty	AF ≤ 1 MHz	AF $\times$ (phase deviation + 1) $\leq 0.3 \times$ demodulation bandwidth					AF $\times$ (phase deviation + 1) $\leq 0.3 \times$ demodulation bandwidth		AF $\times$ (phase deviation + 1) $\leq 0.3 \times$ demodulation bandwidth
		$\pm(0.002 \text{ rad} + 0.002 \times \text{measured value})^{7, 11}$			$\pm(0.02 \text{ rad} + 0.002 \times \text{measured value})^{12}$		$\pm(0.02 \text{ rad} + 0.002 \times \text{measured value})$		0.2 % of reading + residual PM
Residual PM	demodulation bandwidth ≤ 100 kHz, RMS, RF ≤ 1 GHz, highpass 300 Hz								
	0.2 % of reading + residual PM								
		0.3 mrad ⁷			1 mrad		5 mrad		2 mrad

Carrier power versus time		FSW	ESW/ FSMR3000	FSWP	FSVA3000/ FSV3000	FPS	FPL	EPL	RTO
Display range		noise floor to +30 dBm							noise floor to +24 dBm
Measurement uncertainty		unmodulated carrier, S/N > 16 dB							
	RF: 9 kHz to 30 MHz	–	–	–	–	–	–	1 dB	–
	RF: 50 kHz to 3 GHz	–	–	–	1 dB	1 dB	1 dB	–	–
	RF: 9 kHz to 8 GHz	1 dB	1 dB	1 dB	–	–	–	–	–
Maximum dynamic range	< $0.75 \times$ bandwidth of R&S®RTO	–	–	–	–	–	–	–	1 dB
	demodulation bandwidth 200 kHz								
	RF input level $\geq$ (RF attenuation/dB – 10) dBm						RF input level $\geq$ (RF att./dB – 13) dBm		RF input level $\geq$ (RF att./dB – 10) dBm
Nonlinearity of displayed level		90 dB ⁷			75 dB		75 dB		80 dB
	S/N > 16 dB	0.1 dB ⁷			0.2 dB		0.2 dB		1 dB

¹¹ With R&S®VSE and I/Q files the value increases to $\pm(0.002 \text{ rad} + 0.004 \times \text{measured value})$.

¹² With R&S®VSE and I/Q files the value increases to $\pm(0.02 \text{ rad} + 0.004 \times \text{measured value})$.

AF spectrum		
Span		max. $0.5 \times$ demodulation bandwidth
Resolution bandwidth		1 Hz to 10 MHz
RF spectrum		
Span		max. demodulation bandwidth
Resolution bandwidth		1 Hz to 10 MHz
Shape factor	60 dB:3 dB	2.5 (nom.)
Modulation distortion		
Measurement functions		THD, SINAD
Measurement range		–100 dB to 0 dB
Resolution		0.01 dB
Measurement uncertainty		0.5 dB
AF frequency range		10 Hz to 5 MHz
Trigger		
Trigger functions ¹³		RF level ¹⁴ , AM, FM, PM demodulation

¹³ Not available with R&S®VSE.

¹⁴ Not available with FPL.

Ordering information

Designation	Type	Order No.
Analog demodulator options		
AM/FM/PM modulation analysis	R&S®FSW3-KM107	1347.7646.02
AM/FM/PM modulation analysis	R&S®FSW-K7	1313.1339.02
AM/FM/PM modulation analysis ¹	R&S®FSWP-K7	1325.4238.02
AM/FM/PM modulation analysis ²	R&S®FSMR3-K7	1345.3389.02
AM/FM/PM modulation analysis	R&S®FSV3-K7	1330.5022.02
AM/FM/PM modulation analysis	R&S®FPS-K7	1321.4079.02
AM/FM/PM modulation analysis	R&S®FPL1-K7	1323.1731.02
AM/FM/PM modulation analysis	R&S®ESW-K7	1331.6216.02
AM/FM/PM modulation analysis	R&S®VSE-K7	1320.7539.06
Analyzers		
FSWX, FSW, R&S®FSWP, R&S®FSMR3000, R&S®FSVA3000, R&S®FSV3000, FPL		
EMI test receivers		
R&S®EPL, R&S®ESW		
Vector signal explorer		
R&S®VSE basic edition	R&S®VSE	1345.1011.06
R&S®VSE enterprise edition	R&S®VSE Enterprise Edition	1345.1105.06
Software maintenance	R&S®VSE-SWM	1320.7622.81

Service at Rohde & Schwarz
You're in great hands

- ▶ Worldwide
- ▶ Local and personalized
- ▶ Customized and flexible
- ▶ Uncompromising quality
- ▶ Long-term dependability

Rohde & Schwarz

The Rohde & Schwarz technology group is among the trailblazers when it comes to paving the way for a safer and connected world with its leading solutions in test & measurement, technology systems and networks & cybersecurity. Founded more than 90 years ago, the group is a reliable partner for industry and government customers around the globe. The independent company is headquartered in Munich, Germany and has an extensive sales and service network with locations in more than 70 countries.

Sustainable product design

- ▶ Environmental compatibility and eco-footprint
- ▶ Energy efficiency and low emissions
- ▶ Longevity and optimized total cost of ownership

Certified Quality Management
ISO 9001

Certified Environmental Management
ISO 14001

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